

Newborough Warren

What's Happening to the Water Under the Dunes?

A 21-year groundwater study at one of Wales's most important coastal dune sites

Hollingham, M. (2026). Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales. Draft, 2026.

Project: github.com/newbroman/Newborough_Hydrology

Web tools (scenario viewer, flood forecaster): newbroman.github.io/Newborough_Hydrology

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The place

Newborough Warren sits at the southern tip of Anglesey, covering roughly 1,300 hectares of sand dunes, wetlands and conifer plantation. It is designated as a Special Area of Conservation — one of the most important coastal dune systems in Wales.

The site is home to rare plant communities called dune slacks: low-lying hollows between the dunes where groundwater rises close to the surface each winter, creating seasonally flooded wetlands that support creeping willow, orchids and a rich variety of wetland invertebrates.

The survival of these habitats depends entirely on groundwater. If the water table sits too low in summer, the plant communities that define dune slacks shift irreversibly towards drier grassland. The difference between a thriving wet slack and a degraded dry one is remarkably small — just 37 centimetres of water table depth in summer.

A large portion of the northern dunes — around 700 hectares — was planted with Corsican pine between 1948 and 1965. This plantation intercepts roughly a quarter of incoming rainfall before it can reach the ground, reducing the amount of water that recharges the aquifer beneath.

Figure 1: Site Topography and Hydrogeological Features

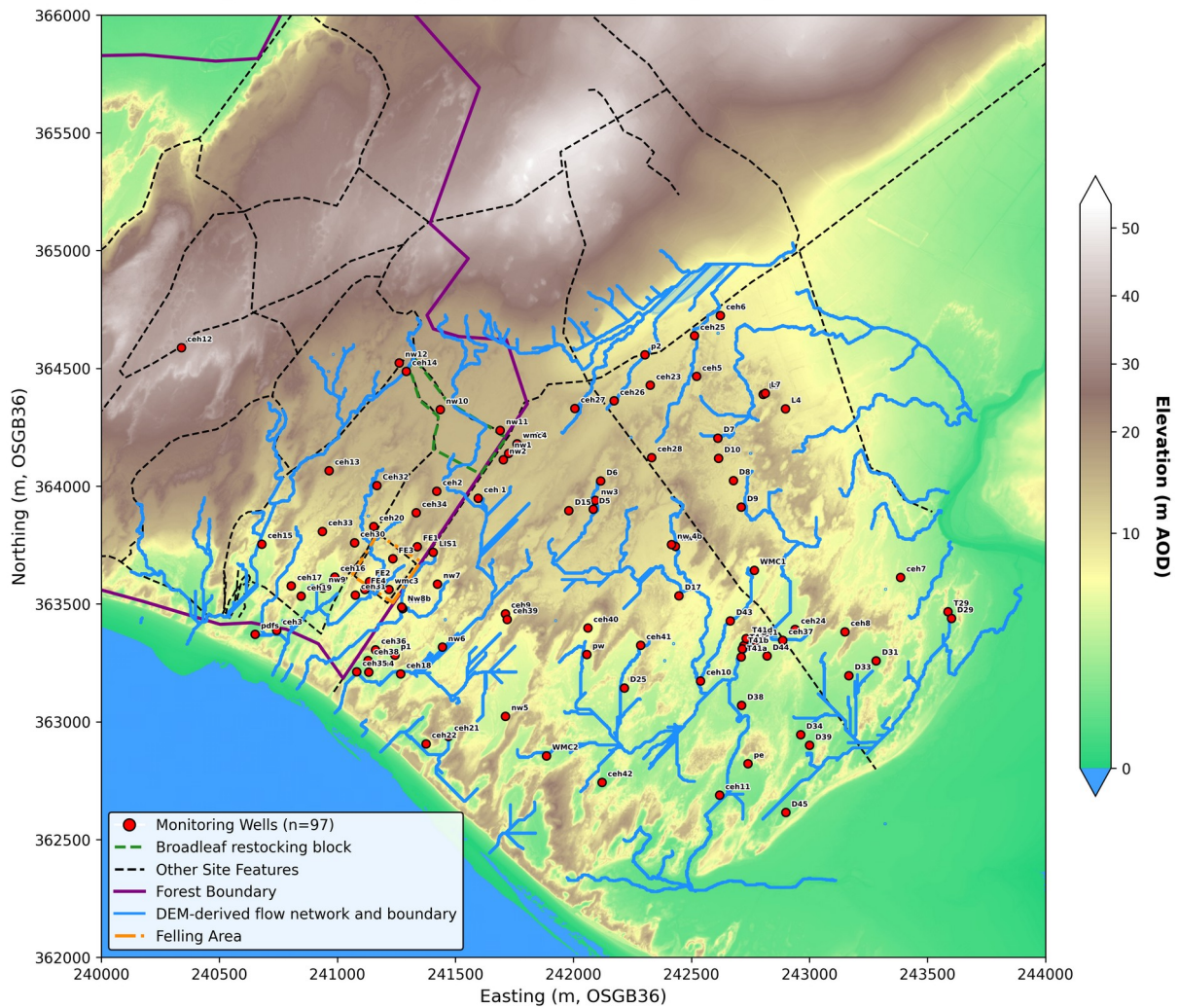


Figure 1. Site overview showing the topography of Newborough Warren, the monitoring network, the forest boundary (magenta), the 2017 felling area (orange), and the drainage network derived from the digital elevation model. The map plots all surveyed dipwell locations; 88 wells passed quality control and entered the analysis.

The study

The study monitored 88 wells across the site for 21 years (2005–2026), measuring water levels monthly with simple hand-held instruments called dipwells. Combined with publicly available rainfall and temperature data from a nearby weather station, this network was analysed using a 31-step data pipeline to understand how the groundwater system works, how it is changing, and what management can realistically achieve.

The approach was deliberately designed to be low-cost and reproducible — no specialist equipment, no telemetry, no laboratory analysis — so that it could be replicated at other coastal dune sites facing similar challenges.

How the model works

At the heart of the analysis is a simple water balance equation fitted independently at each well. Each month, the change in water level is explained by three competing forces: rainfall pushing the water table up, evaporation pulling it down, and drainage carrying water sideways towards the coast or lake. The model estimates the strength of each force for every well, producing three coefficients that together define how that part of the aquifer behaves. Wells with similar coefficients are grouped into zones.

This state-space model (SSM) was benchmarked against a simpler transfer function approach that lacks the drainage feedback term. Figure 2 shows both models against observed water levels at CEH6, a Dune cluster well. In one-step diagnostic mode (top panel) the two models are comparable, but in iterative forecasting mode (bottom panel) — where the model runs forward using only climate data and its own previous predictions — the SSM maintains realistic seasonal cycles (NSE = 0.66) while the transfer function diverges (NSE = −1.1). Across the network the SSM gave a positive forecasting score at 65 of 66 wells, against 43 of 66 for the simpler model. The drainage term is what prevents the model from drifting.

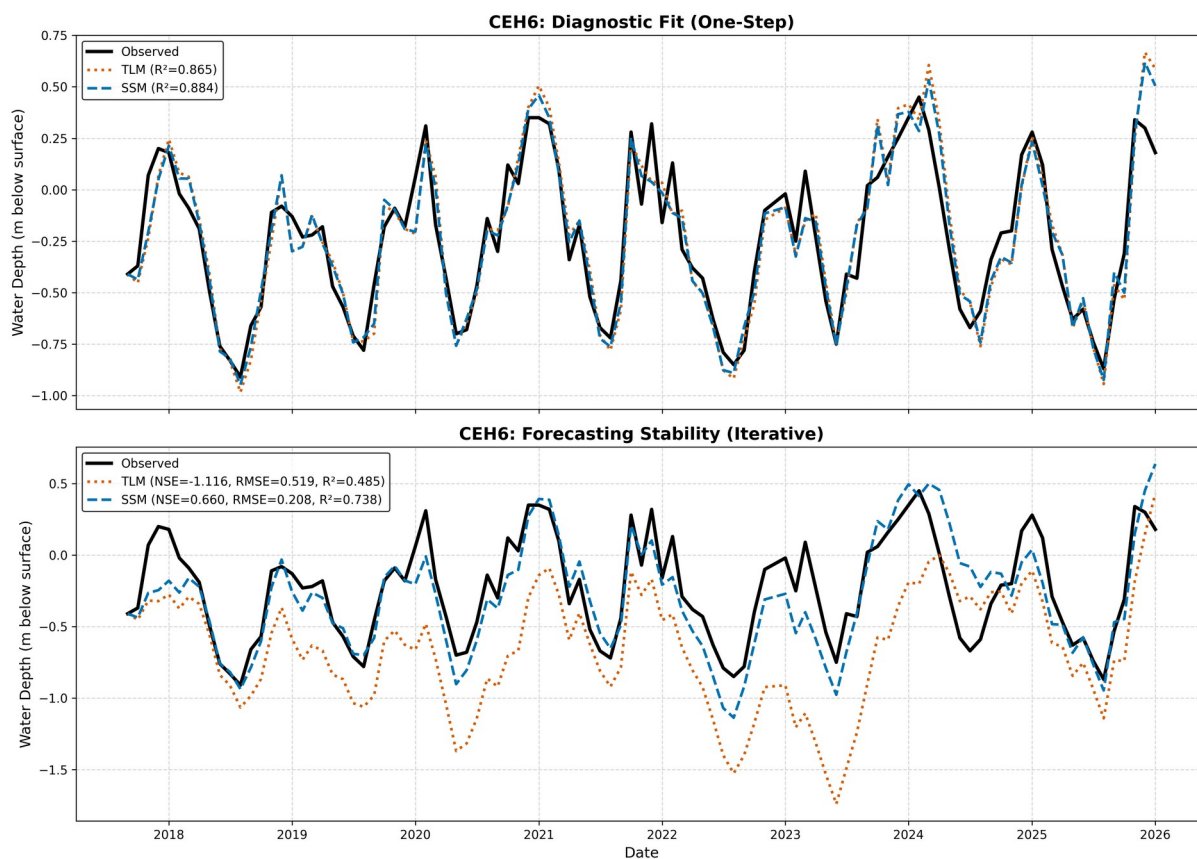


Figure 2. Model benchmarking at CEH6 (Dune cluster). Top: one-step diagnostic fit, where both the SSM (blue dashed) and transfer function (orange dotted) track the observed water level (black). Bottom: iterative forecasting, where the model runs forward using only climate forcing. The SSM maintains realistic amplitude and timing; the simpler model diverges progressively.

Seasonal prediction equations

Alongside the monthly SSM, a pair of seasonal prediction equations was fitted for each groundwater zone. A winter equation predicts the peak winter water level from cumulative winter rainfall (October–March) and the preceding summer minimum. A summer equation predicts the summer minimum from cumulative summer rainfall (April–September) and the preceding winter peak. Together these capture the aquifer’s memory: how deep the summer drought was determines how much winter rain is needed to refill, and how high the winter peak reached determines how far the water table can fall the following summer.

For example, the Dune zone (C2) summer equation is:

$$\text{Summer minimum} = 0.0018 \times P_{\text{summer}} + 0.54 \times h_{\text{winter peak}} - 1.65 \quad (R^2 = 0.65)$$

This says: for every additional 100 mm of summer rain, the summer minimum is about 0.18 m shallower; and the higher the preceding winter peak, the shallower the following summer minimum. The intercept (–1.65 m) represents the structural deficit — even with average rainfall and an average winter peak, the summer minimum sits well below the ground surface.

Critical rainfall thresholds

Because the SSM is a physical mass balance rather than a statistical correlation, it can be run forwards: given a measured summer water level and a forecast of winter rainfall, the model calculates whether the water table will rise high enough to flood a given slack. This produces a critical rainfall threshold for each well — the amount of winter rain needed to achieve flooding, expressed as a multiple of the long-term average. Wells that need roughly twice the average winter rainfall or more are classified as structurally unreachable under any winter in the long climate record.

These equations are built into an interactive web tool (the scenario viewer) that allows site managers to explore how different management actions — tree felling, thinning, broadleaf conversion, dune scraping — and future climate scenarios would shift water levels across the entire site. A companion flood forecaster converts a single dipwell reading into a probability of winter flooding, updated as the season progresses.

Five distinct groundwater zones

The analysis identified five distinct zones across the site, each behaving differently depending on its geology, position and land cover:

- **C1 — Lake Edge (eastern block):** Shallow water table, responds quickly to rainfall, drains fast. The closest to the ecological tipping point.
- **C2 — Dune (eastern block):** Classic open dune, moderate rainfall response.
- **C3 — Western Residual:** Acts as a deep buffer — slower to respond, holds water longer.
- **C4 — Main Forest:** Pine plantation on thin soil over bedrock. Trees intercept rain and the shallow substrate amplifies summer drying.
- **C5 — Coastal Forest:** Pine plantation on deeper coastal sand. Shows the steepest water table decline of any zone — declining more than three times faster than the next worst.

A key finding is that it is the thickness of sand beneath each area — not the trees above — that controls how severely evaporation draws down the water table in summer. Within the forest zone, ground elevation alone explains about 95% of the variation in this drying effect.

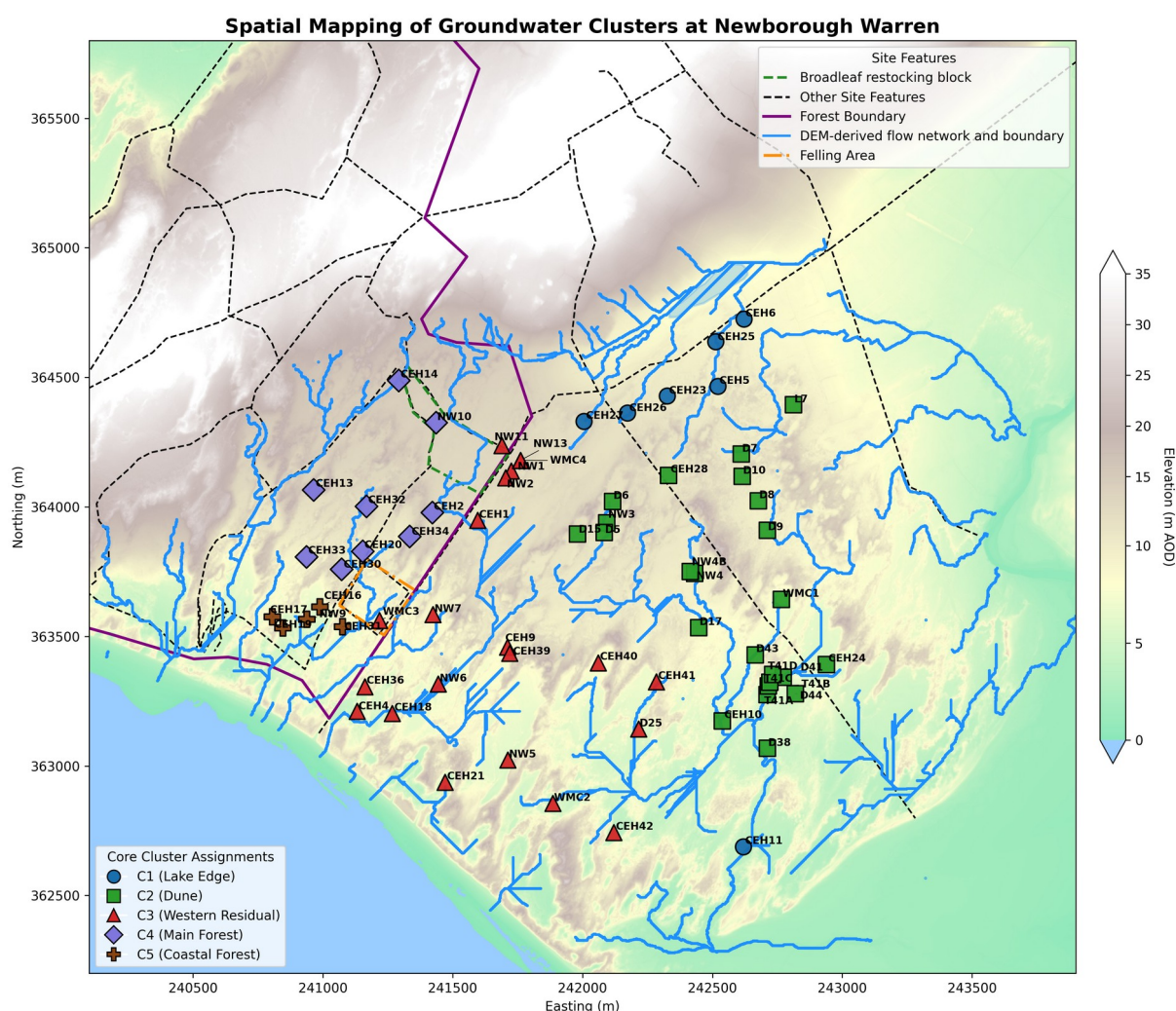


Figure 3. Spatial mapping of the five groundwater clusters identified by the analysis. Each colour represents a zone with distinct behaviour: C1 Lake Edge (blue), C2 Dune (green), C3 Western Residual (red), C4 Main Forest (purple), C5 Coastal Forest (brown).

How the five zones compare

The table below summarises the key characteristics of each groundwater zone. The ‘recharge sensitivity’ measures how strongly rainfall raises the water table; the ‘evaporation draw’ measures how strongly summer heat lowers it; the ‘drainage rate’ measures how quickly water drains away; and the ‘canopy share’ measures how much of the rainfall energy budget is consumed by evaporation from the canopy and soil rather than reaching the water table. The first three coefficients are dimensionless model parameters — larger values mean a stronger effect.

Zone	Character	Recharge sensitivity	Evaporation draw	Drainage rate (per month)	Canopy share
C1 Lake Edge	Shallow, fast-draining, near tipping point	4.58 (highest)	0.96 (lowest)	0.090 (fastest)	22%
C2 Dune	Open mature dune, moderate response	3.98	1.77	0.066	26%
C3 Western Residual	Deep buffer, slow response	3.57	1.85	0.058	28%
C4 Main Forest	Pine on thin substrate over bedrock	2.52 (lowest)	2.55 (highest)	0.020 (slowest)	40%
C5 Coastal Forest	Pine on deeper coastal sand; steepest decline	2.41	1.32	0.045	41%

Table 1. Summary of the five groundwater zones. Higher recharge sensitivity means a stronger water table response to rainfall; higher evaporation draw means more severe summer drying; higher canopy share means more rainfall intercepted by trees.

The central finding: summer is what matters

Conventional thinking assumes that winter rainfall is what fills dune slacks. This study found the opposite: it is the summer minimum water level that determines whether a slack will flood the following winter. If the water table drops too low during summer, no realistic amount of winter rain can refill the deficit.

This matters because summers are getting warmer. Since 2013, summer temperatures at the site have been nearly 1°C above the long-term average, increasing evaporation and deepening the summer drought.

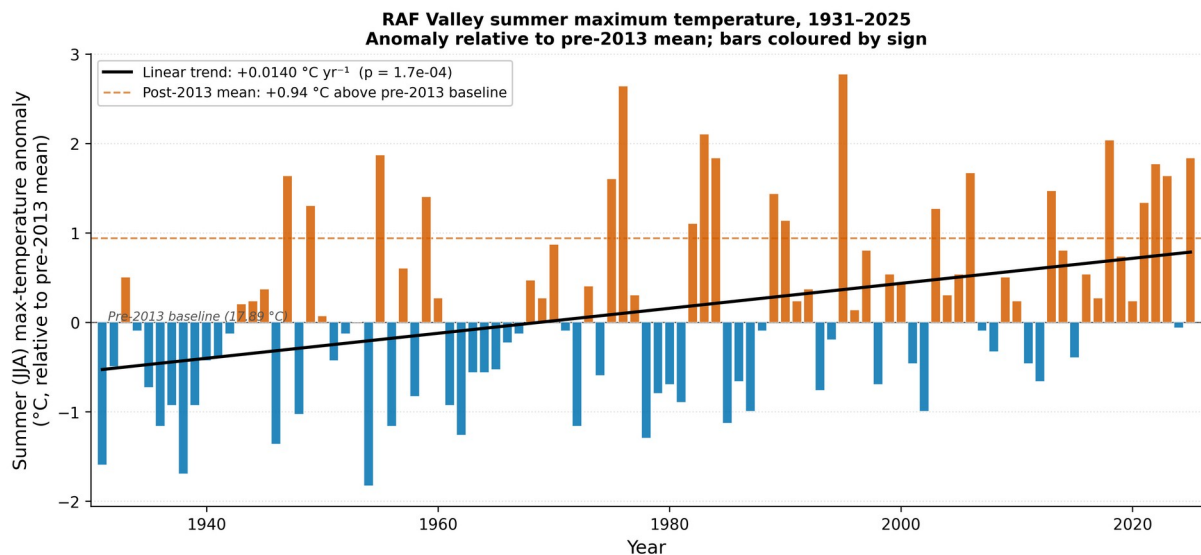


Figure 4. Summer temperature anomaly at RAF Valley (nearest weather station), 1931–2025. Orange bars show years warmer than the pre-2013 average; blue bars show cooler years. The dashed orange line marks the post-2013 mean: +0.94°C above baseline.

The water table has been declining across the network, with the trend statistically clearest at the Lake Edge (C1) and Coastal Forest (C5) zones; the Dune (C2) trend is marginal, and the Western Residual (C3) and Main Forest (C4) trends are not statistically significant on their own. Trend-line extrapolation indicates that the Lake Edge zone — where many of the most ecologically valuable slacks are located — is on course to cross the threshold for viable dune slack habitat around 2030–2032. Individual dry summers are already reaching that threshold.

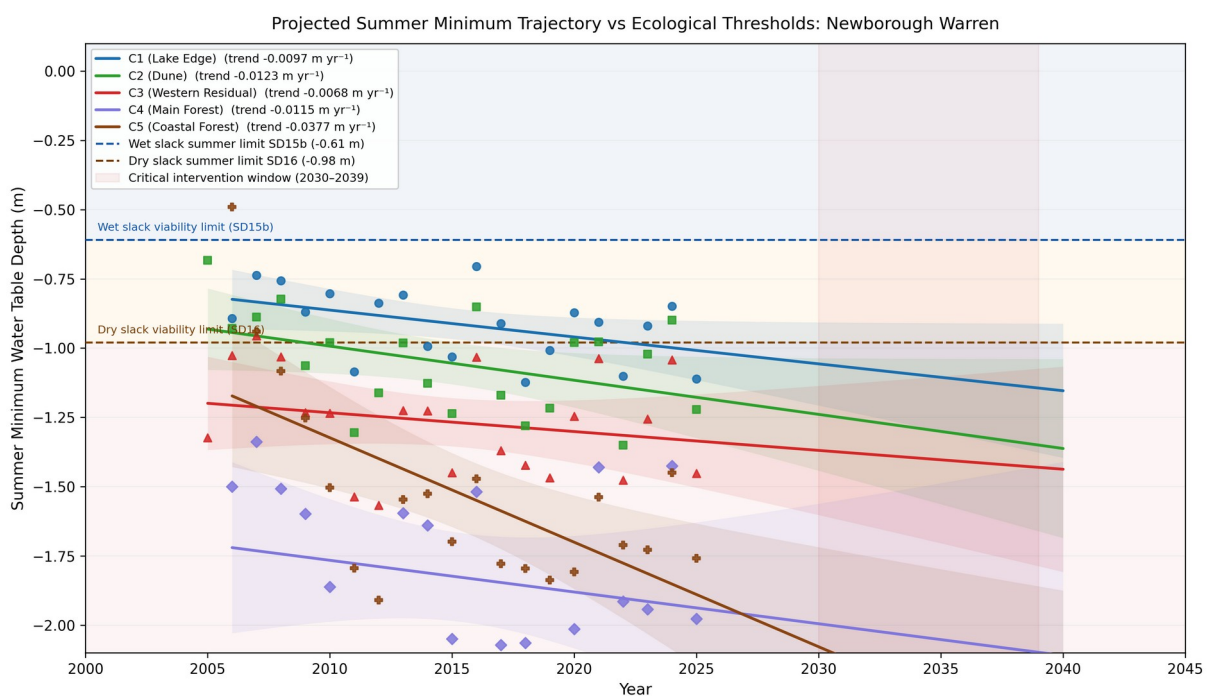


Figure 5. Projected summer minimum water table depth for all five clusters. The blue dashed line marks the wet slack threshold (−0.61 m); the brown dashed line marks the dry slack threshold (−0.98 m). The shaded band highlights the critical intervention window (2030–2039). All clusters are tracking downward.

A second measure: the spring baseline

The summer minimum is not the only way to read the health of a dune slack. A second measure — the five-year mean spring water level, or MSL5 — describes the sustained baseline within which slack vegetation actually establishes itself, averaged over the spring

months and smoothed across five years to filter out single wet or dry seasons. Recent ecological work (van Willegen et al. 2025) identifies this spring baseline as the single best predictor of how the plant communities respond, which makes it a valuable companion to the summer minimum.

The two measures tell noticeably different stories. While the summer minimum is deepening sharply, the spring baseline is far more stable: it is projected to shift by only about 21–39 mm under the 2080s climate trajectory — three to five times less than the parallel deepening of the summer minimum. As a result the spring baseline stays within its present ecological regime at every zone, with the open-dune zones (C1, C2) sitting comfortably in the wet-slack range throughout the record and only the Coastal Forest (C5) sitting below the dry-slack threshold. In other words, it is specifically the summer low point — the few weeks when the water table bottoms out — that is breaching the ecological thresholds, not the year-round baseline.

This is why the summer minimum is the binding constraint, and why interventions must be judged on their summer effect rather than on annual or spring-average measures, which can look reassuring while the critical summer low point continues to fall. In this study only dune scraping registered a benefit on both measures; the recovery after tree felling did not show up on either in the way that matters.

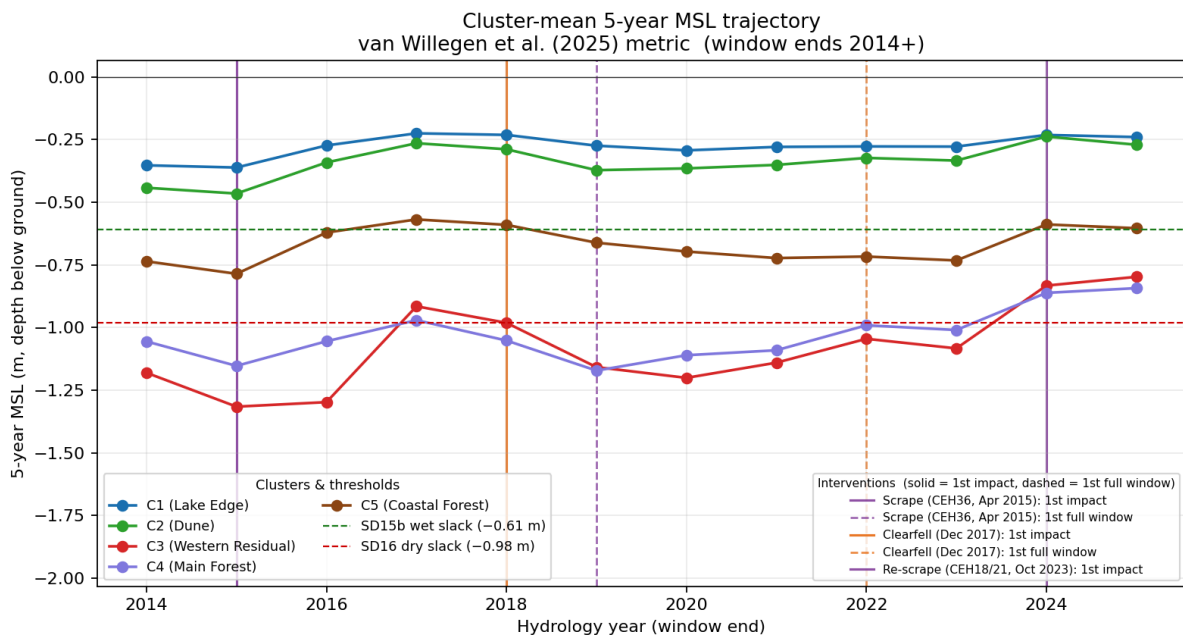


Figure 6. The five-year mean spring water level (MSL5) for each zone, 2014–2025. Compared with the summer minimum (Figure 5), the spring baseline is markedly flatter and more stable. The open-dune zones C1 (blue) and C2 (green) sit above the wet-slack threshold (green dashed); only the Coastal Forest (C5, brown) and parts of the western zones approach the dry-slack threshold (red dashed). Vertical lines mark the scraping and felling interventions.

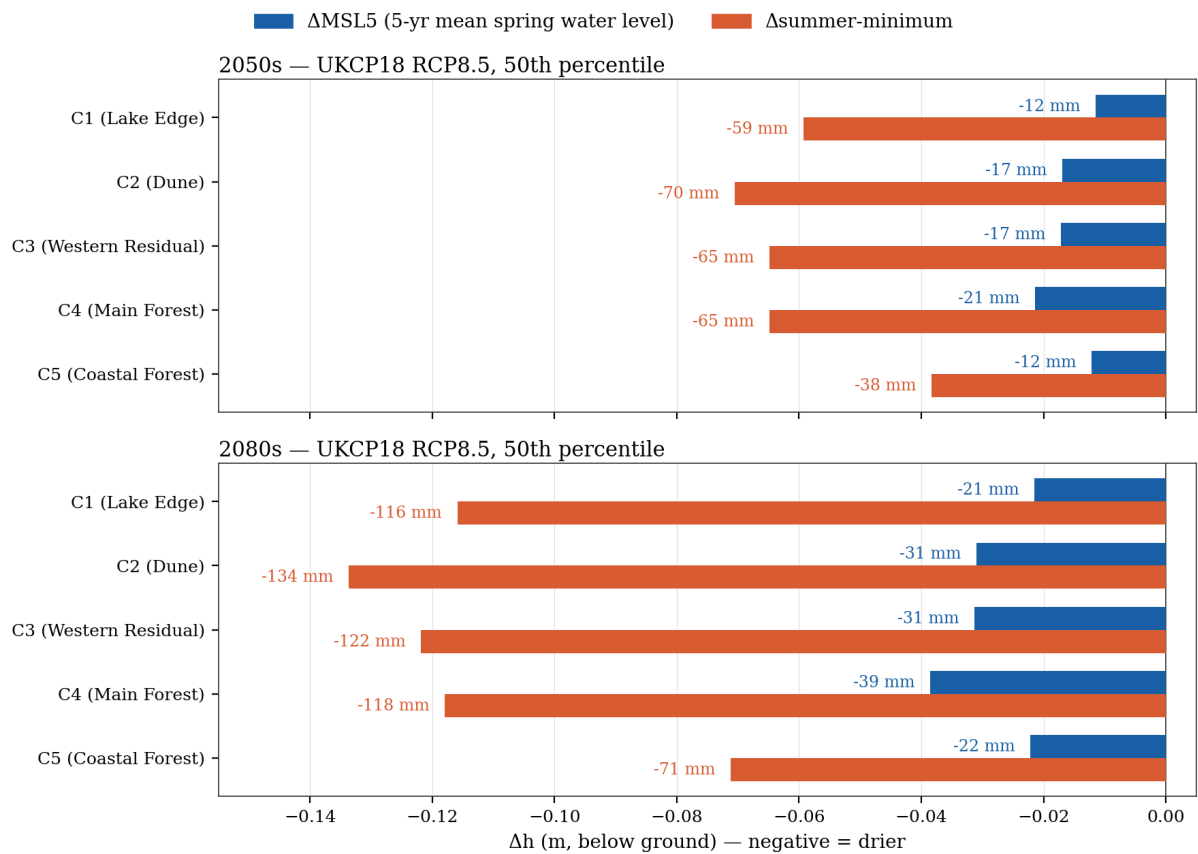


Figure 7. The projected drying of each zone by the 2050s and 2080s (UKCP18 high-emissions scenario), comparing the two ecological measures side by side. Blue bars show the shift in the spring baseline (MSL5); orange bars show the shift in the summer minimum. Bars extend left for a drier (deeper) water table. At every zone the summer minimum falls several times further than the spring baseline — by the 2080s the spring baseline drops by 21–39 mm while the summer minimum drops by 71–134 mm. It is the summer low point, not the year-round baseline, that bears the brunt of the projected climate change.

What management has been tried

Dune scraping

Dune scraping involves mechanically removing the top layer of sand to bring the ground surface closer to the water table. At one well (CEH36), three independent methods converged on a scraping benefit of roughly 80–140 mm, and the paired summer minimum shift against an unscraped control well was +195 mm ($p = 0.004$) — more than half of the critical 37 cm ecological gradient.

However, scraping only works where it is placed correctly within the landscape. At more coastal locations (CEH18 and CEH21), the benefit was overwhelmed by a separate problem: the coastline itself is retreating, which lowers the water table from the seaward edge inward.

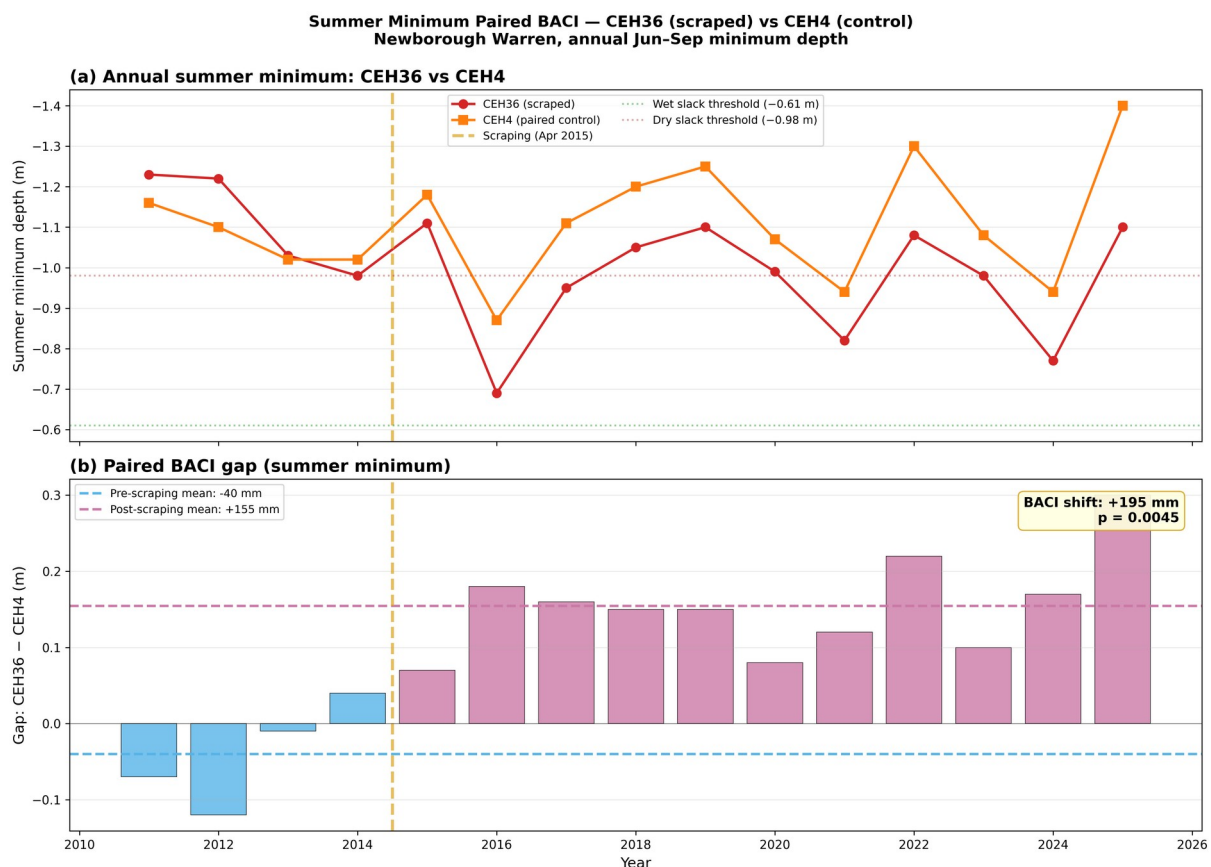


Figure 8. Summer minimum depth at the scraped well CEH36 (red) versus its paired control CEH4 (orange). Top: the two wells track together before scraping in 2015, then CEH36 sits consistently shallower. Bottom: the gap between them shifts from -40 mm before scraping to $+155$ mm after, a paired BACI shift of $+195$ mm ($p = 0.004$).

The reason scraping works at CEH36 is worth being precise about, because it sets a limit on what scraping can achieve elsewhere. The benefit is geometric: lowering the ground surface permanently moves it closer to the water table, so the same water table sits relatively higher beneath the slack. Scraping does not add water to the aquifer. This distinction matters, because the obvious alternative — trying to raise the water table itself by capturing more winter rainfall — runs into a hard physical constraint. The extra recharge steepens the local hydraulic gradient and drains away through the highly transmissive sand before the summer minimum is reached, so it provides little benefit at the time of year that determines whether a slack survives.

Scraping is therefore best understood as a targeted, local re-profiling of the ground rather than a way to recharge the aquifer, and its value is greatest at inland transitional sites in the C1 and C2 zones where the aquifer base is stable and the coastline is not yet drawing the

water table down. Even there, the benefit is gradually eroded by the background climate trend, so scraping should be seen as buying time at carefully chosen locations rather than as a site-wide solution.

Tree felling

In December 2017, approximately 8.4 hectares of Corsican pine were clearfelled as a management experiment. The study used a rigorous five-tier experimental design with 17 wells and three independent control groups to test whether removing the trees raised the water table.

The result was nuanced. Comparing felled wells against unfelled forest — the most direct test — the analysis detected a statistically significant rise in mean monthly water levels of around +120 mm at the core impact well ($p < 0.001$). The rise of +33 mm at the forest edge was not statistically significant ($p = 0.19$), and a synthetic extension of within-compartment wells returned a step of +85 mm ($p < 0.001$). Crucially, the summer minimum depths — the metric that determines whether dune slacks survive — showed no corresponding improvement: felled and unfelled wells tracked each other through the summer. The clearfell raised average water levels but could not overcome the concurrent loss of recharge efficiency during the summer months when it matters most.

The analysis suggests the forest canopy acts as both a sink (intercepting rain) and a buffer (shielding the ground from direct evaporation in summer). Removing the canopy recovers some winter recharge but simultaneously exposes the soil to greater summer drying, and the two effects roughly cancel out at the time of year that matters — a point explored further below in the context of the site-wide decline in recharge efficiency.

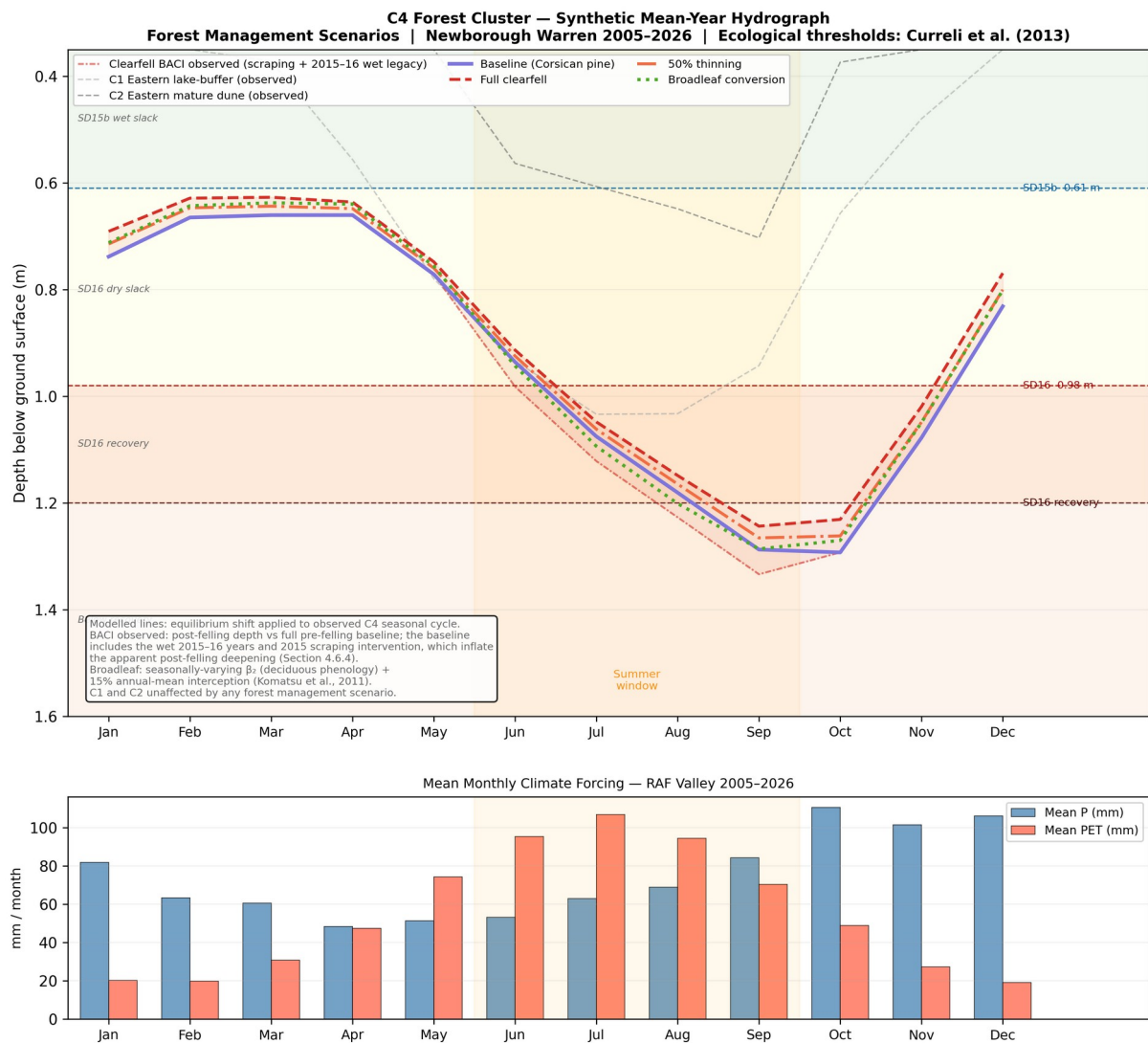


Figure 9. The seasonal water table cycle in the Main Forest zone (C4), showing the current baseline (blue) alongside modelled scenarios for full clearfell (red dashed), 50% thinning (orange) and broadleaf conversion (green dotted). The ecological threshold bands are shown as coloured horizontal zones. The bottom panel shows the seasonal pattern of rainfall versus evaporation.

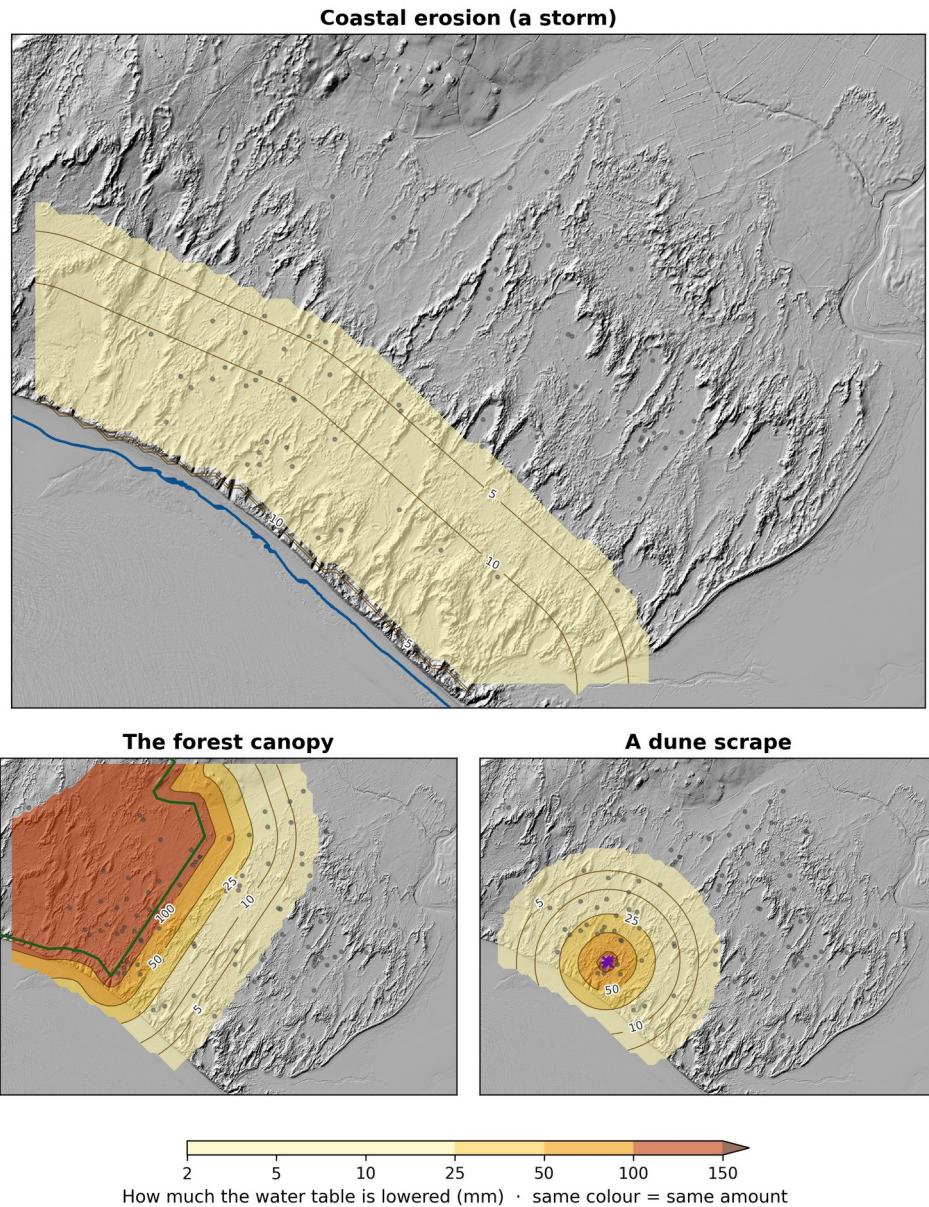
Crucially, changes to the forest do not propagate sideways to the open dune slacks where the ecological need is greatest. The forest and the open dunes operate as largely independent groundwater systems, and the reason is the site's drainage geography: water beneath the forest flows south-westward towards Caernarfon Bay, while the open eastern dunes drain south-eastward towards the Menai Strait, with a topographic high separating the two. The forest's drawdown footprint, modelled separately, drops from over 50 mm within about 100 m of the forest edge to 1 mm or less at the eastern Lake Edge and Dune wells where the most valuable slacks lie. Forest management changes are carried away down the forest's own drainage path and never reach the eastern slacks.

Three drivers, three different footprints

It helps to see the three local drivers of water-table lowering on the same map and the same scale. Coastal erosion (illustrated as a single storm event) acts as a thin band along the shore but covers a large area. The forest canopy acts broadly and continuously, with a wide footprint concentrated around the plantation. A dune scrape acts intensely but only locally — large near the cut and falling away rapidly with distance. Crucially, none of them is a switch that can simply be flipped to reverse the others: the forest cannot push back coastal retreat,

scrapping cannot recover canopy interception, and the climate-driven background trend reaches across all three. The drivers overlap in some places and miss each other entirely in others, which is why management has to be planned at the level of specific slacks rather than the site as a whole.

Several things lower the water table at Newborough Warren



Each driver has a different size and reach: the forest acts broadly, a scrape intensely but only locally, coastal erosion along the shore. Changing any one has knock-on effects elsewhere — there is no single switch.

Figure 10. The three local drivers of water-table lowering shown together on a shared colour scale (same colour = same amount of drawdown, in millimetres). Top: coastal erosion from a single storm, spread along the shore. Bottom left: the forest canopy, acting broadly within and just outside the plantation. Bottom right: a dune scrape, intense but local. Each driver has a different size, shape and reach — there is no single intervention that addresses all three.

A declining recharge efficiency

Perhaps the most concerning finding is a site-wide decline in recharge efficiency — the proportion of rainfall that actually reaches the water table. This decline is the strongest candidate mechanism for the observed summer minimum deterioration across the entire monitoring network. It affects all five zones, regardless of whether they are forested or open, scraped or untouched, and operates independently of canopy cover. It is consistent with changing rainfall patterns (fewer sustained soaking events, more intense but brief storms that run off rather than infiltrating) and represents a climate-driven trend that no on-site management intervention can address.

This same trend explains why both interventions fall short in summer. When the trees were felled, the canopy stopped intercepting rain and winter recharge did increase — but that extra water steepened the local hydraulic gradient and drained away through the transmissive sand before the summer minimum arrived, just as it does after scraping. At the same time, removing the canopy exposed the soil to greater summer drying. The two effects roughly cancel out, so the felled wells track the unfelled forest through the summer. In the scenario modelling, clearfell and thinning produce only small positive net annual responses at the forest clusters (clearfell about +8 to +10 mm per month water-equivalent), while broadleaf conversion is close to neutral — all far smaller than the climate-driven changes. The common thread is that neither scraping nor felling can hold winter water in the aquifer until summer; the water table's summer low point is set by the site-wide decline in recharge efficiency, not by what is done to the surface or the canopy above it.

Coastal erosion: a hidden and lagged threat

The coastline at Newborough is retreating. As the beach erodes, the aquifer loses volume at its seaward edge, and the water table adjusts downward. A network-scale analysis attributes roughly 41% of the Coastal Forest zone's exceptional decline to this coastal-retreat signal, with the fitted effect reaching some hundreds of metres inland from the eroding western shoreline. Because groundwater moves slowly through sand, the effects of coastal erosion take years to propagate inland, so the decline visible in today's monitoring data partly reflects erosion that happened years ago. If erosion is accelerating, the worst effects have not yet arrived at interior wells.

Two control wells at different distances from the coast are both deteriorating in a pattern consistent with progressive coastal retreat, independently of any management action. Coastal-margin slacks may eventually require managed retreat rather than repeated restoration.

What this means for the future

The study's conclusions are sobering but clear:

1. **The window for effective intervention is finite** — measured in roughly one to two decades at the most vulnerable sites.
2. **Targeted dune scraping at carefully chosen inland sites** is the most effective direct intervention available, but its benefits are eroded over time by the background climate trend.
3. **Tree felling raises average water tables in the forest but does not improve summer minima** — the metric that determines dune slack survival. The evidence supports retaining forest cover while managing canopy density to partially recover the rainfall interception penalty without losing the summer buffering benefit. Forest management does not propagate to the open dune areas where habitat is most threatened.
4. **Climate change is the binding constraint.** Rising summer temperatures, declining recharge efficiency and coastal erosion are all operating in the same direction, and none can be addressed by on-site habitat management alone.
5. **Continued monitoring is essential.** The predictive tools developed in this study — including critical rainfall threshold equations and an interactive forecasting system — give site managers the ability to track whether conditions are on course to cross ecological thresholds, and to prioritise interventions where they can still make a difference.